AI-Enhanced Receipt Management System

Member: Huang Ziwen , Chen Zhi

Team: TOEFL

Course: Software Engineering

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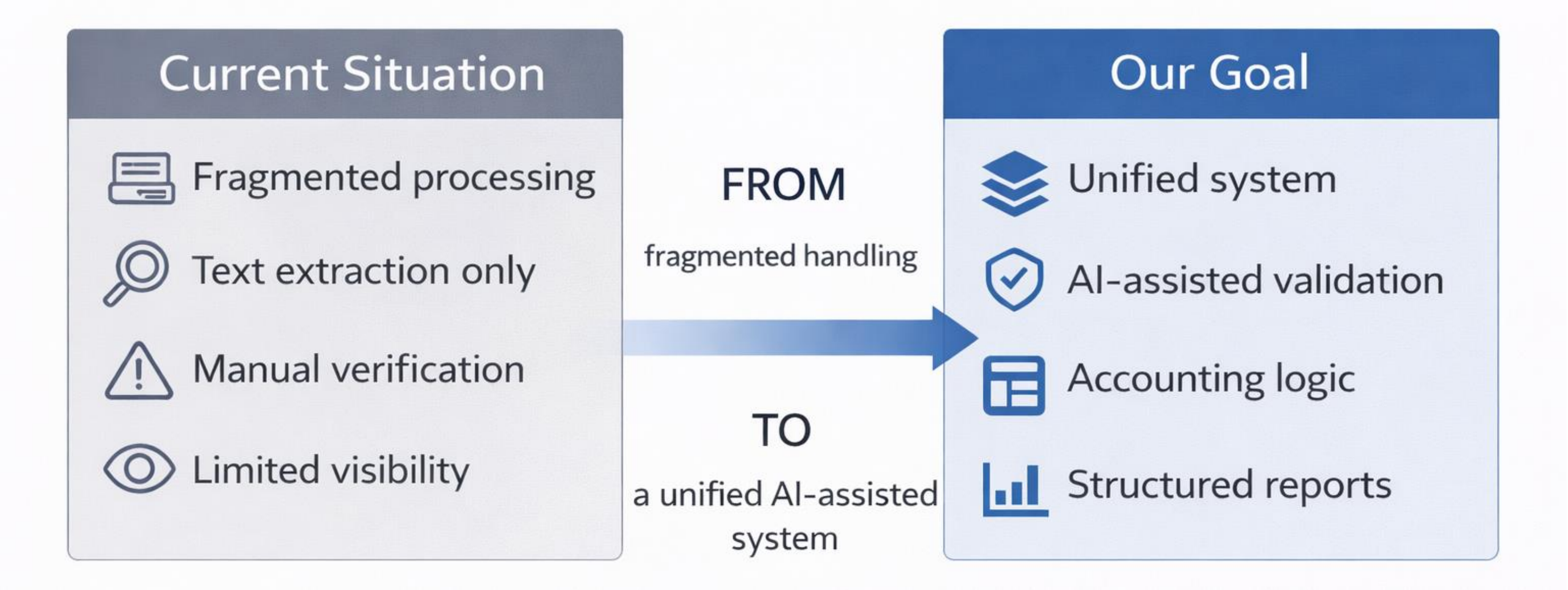
1 System Design

System Design



Problem Context

•Expense and receipt processing is still **fragmented and manual**

•OCR tools extract text, but **do not support validation or accounting logic**

•Small teams lack **lightweight, policy-aware** financial systems

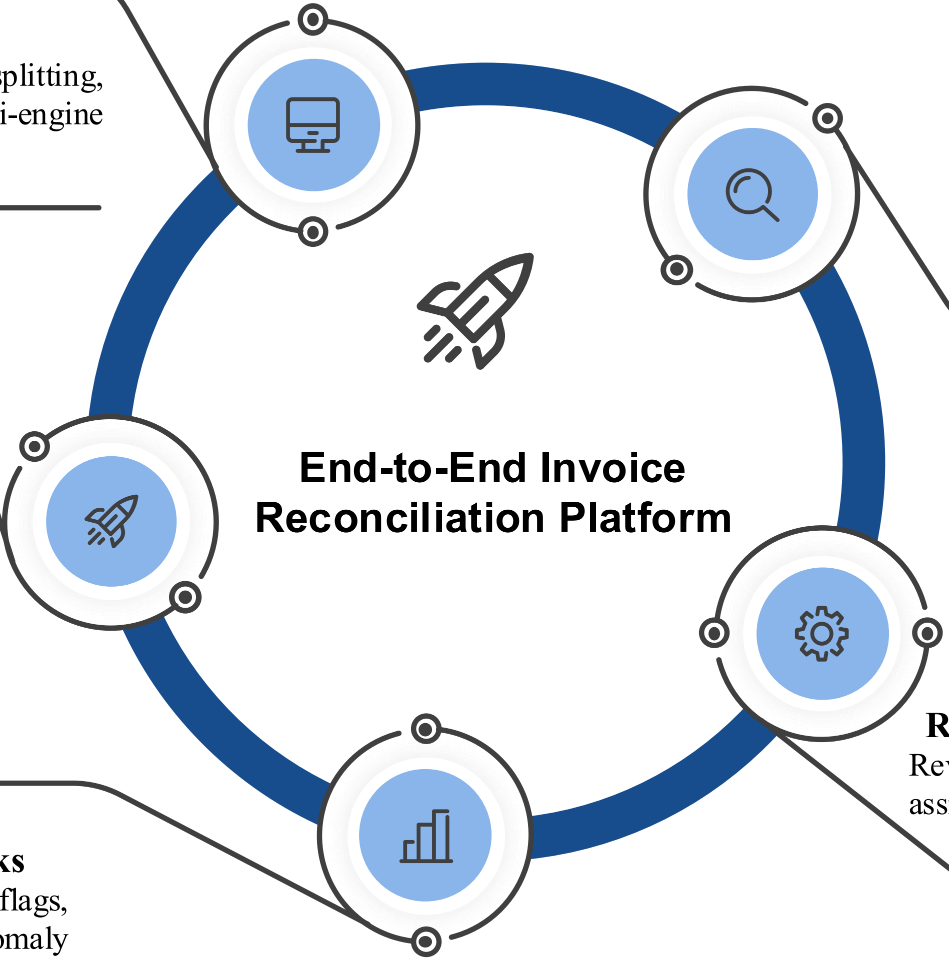
Project Goal

•Design a **complete AI-assisted financial workflow**

•Support **ingestion, validation, accounting, and reporting**

•Emphasize **system integration and reliability** over individual models

System Design



01

Ingestion & OCR

Multi-format upload, PDF splitting, image normalization, multi-engine OCR

02

LLM Reasoning & Generation

Schema-aligned field extraction and result generation with robust fallback mechanisms

03

Policy & Anomaly Checks

Policy-aware validation, risk flags, duplicate and statistical anomaly detection

04

Journal & Vouchers

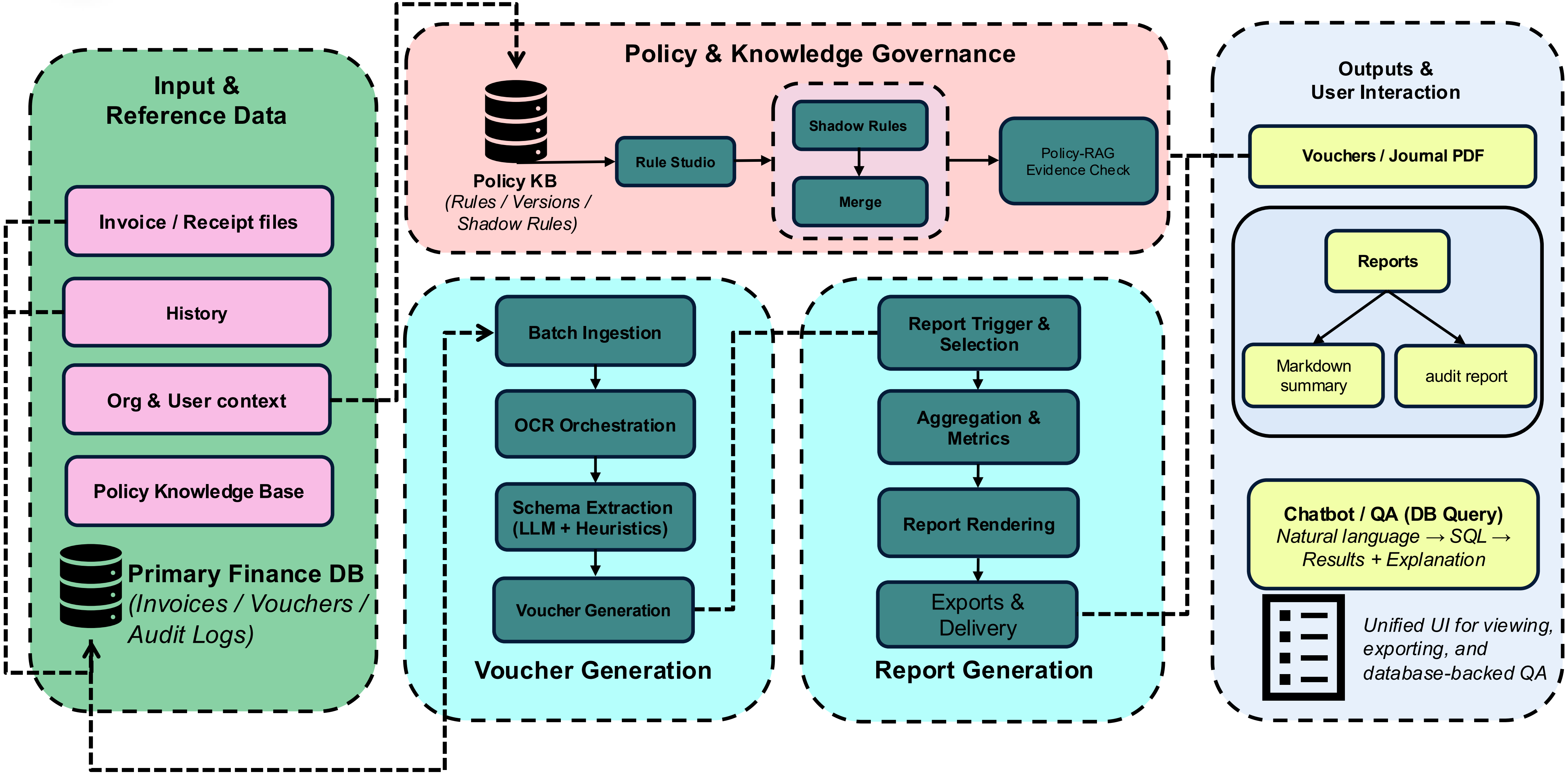
Automatic debit–credit entries and exportable accounting vouchers

05

Reports, Dashboard & AI Assistant

Review reports, visual dashboards, and AI-assisted audit decision support

System Design



System Design(components)

Perception Layer

Document Layout Analysis

- Layout structure detection
(tables / stamps / QR codes; *CV-based seal localization*)
- **Scan-type detection & orientation**
(portrait / landscape; *CV-based rotation/deskew cues*)

Multi-Engine OCR

- Baidu invoice OCR + general OCR
- Text output with confidence score

Understanding Layer

Structured Field Extraction (NER/IE)

- LLM extraction: invoice ID / vendor / amount / date
- Regex fallback: amount & date normalization

Intelligent Classification (Classification)

- Expense category: travel / equipment / office, etc.
(rules + LLM)
- Cost center: auto assignment to department / project

Vectorization (Embeddings)

- SentenceTransformer Chinese embeddings
- Foundation for retrieval and clustering

System Design (components)

Decision Layer

RAG Policy Retrieval

· Supports shadow rules (gradual rollout)

· Outputs compliance flags (**PolicyFlag**)

Retrieve → summarize → semantic match → two-stage LLM validation

Anomaly Detection

· Duplicate reimbursement (vendor + amount + date)

· Amount outliers: Z-score / MAD / IsolationForest

Vendor behavior comparison

Understanding Layer

Conversational Assistant (LLM Chat)

· Finance analyst: data aggregation + natural-language answers

· Smart Q&A: Text-to-SQL

Report Generation (Summarization)

· Review report: risk summary + policy reminders

· Decision summary: trend analysis + attribution insights

Intent Detection (Intent Detection)

· Understand user questions → generate a query plan

· Decide whether additional information is needed

2 Technical Highlights

Technical Highlight

- **Finding:** In real-world invoice review, off-the-shelf AI API calls alone are not enough due to dynamic policies, noisy documents, and edge cases (duplicates/outliers) that cause unstable outputs and false alarms.

- **Prospect:** To make the system more **template-driven, automated, and reliable**, we designed it from both **architecture and algorithm** perspectives. The next slides as two representative highlights showcase these key engineering decisions, while similar optimizations are applied across other modules as well.

Technical Highlight

Retrieval-Augmented Generation (RAG) Architecture

Challenges

Policy drift & versioning

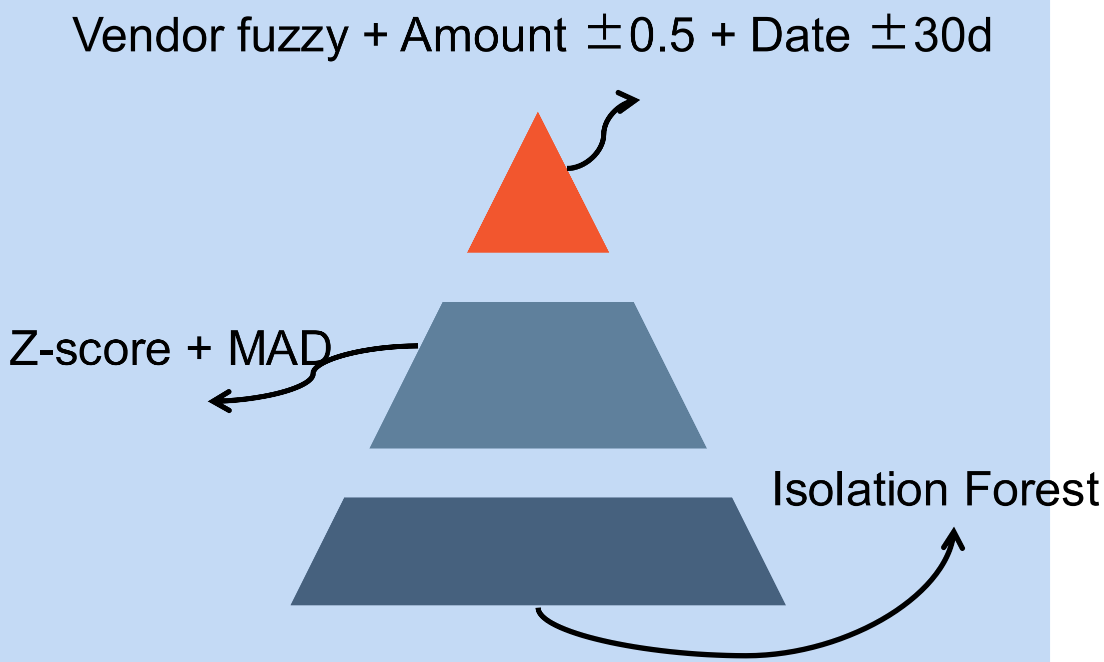
rules change often, hard-coded logic breaks quickly

Hallucination vs auditability
pure LLM lacks traceable evidence for compliance decisions

Core Highlight

- **120-char summaries→+20%** retrieval accuracy (vs full-text embedding)
- **Query rewrite (on/off)** → better match to policy language
- **Two-stage LLM** → decision+rationale → **PolicyFlag JSON**
- **Shadow rules** → prod vs canary diff for human review

Technical Highlight



fuzzy duplicate matching + Z-score/MAD + Isolation Forest

Multi-Algorithm Anomaly Detection

Problem: Single detectors miss different anomaly patterns, so we use an ensemble to reduce misses and false alarms.

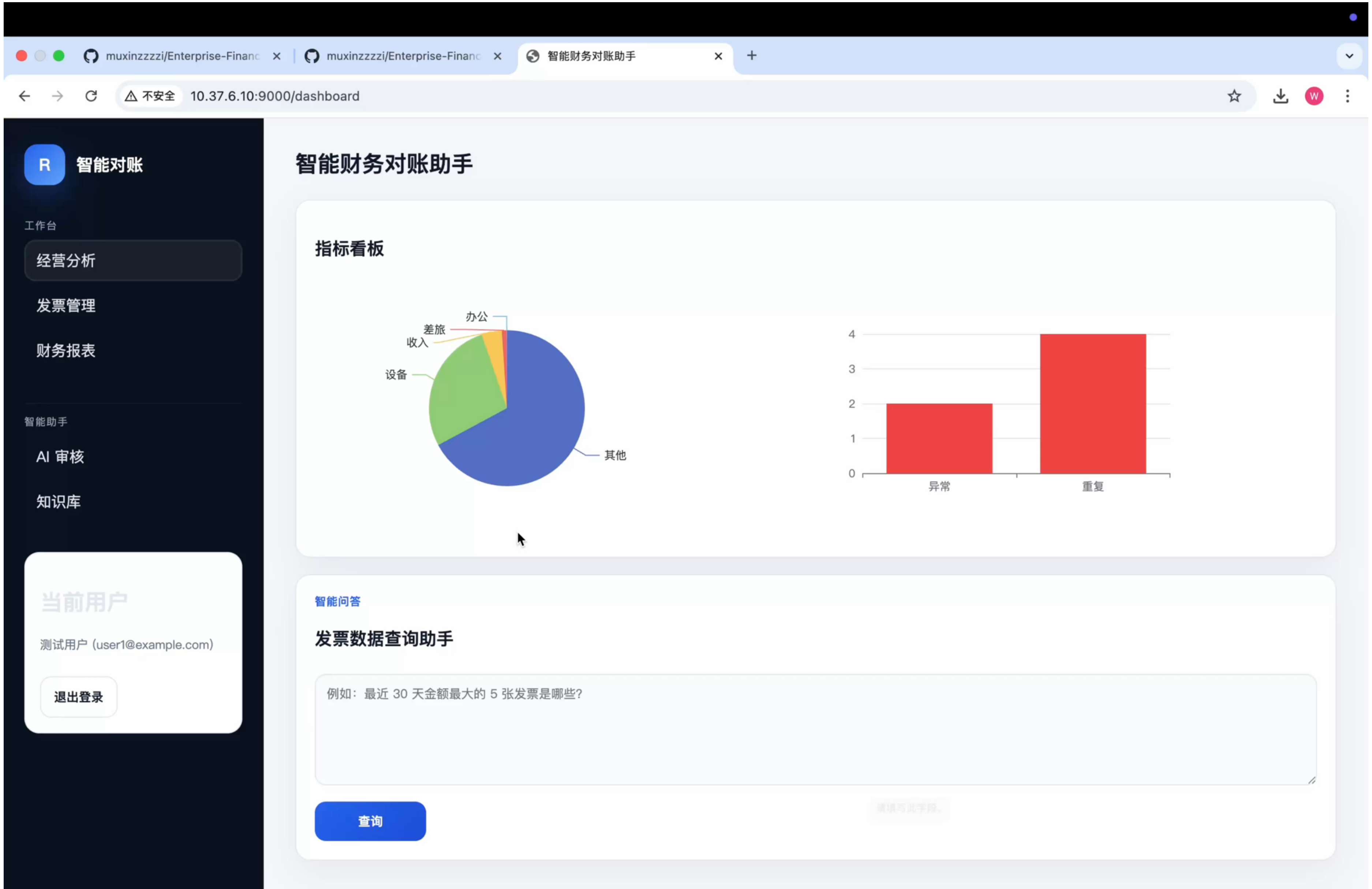
Key Engineering Decision: We combine fuzzy duplicate matching (vendor + amount ±0.5 + date ±30d) with Z-score/MAD and Isolation Forest to output a final anomaly flag.

Method	Recall	FPR
Single	68–79%	38%
Ensemble	**95%**	**12%**

3 Demo & Evaluation

Demo & Evaluation

Demo



Demo & Evaluation


发票号码: 56256956
手写无效
日期 2017•9•9
代码 0000017206
车号 65-51286
上/下车 15•56-16•07
单价 1•90 元
里程 3•8 Km
等候 0•00
金额 10•00 元
燃油附加 1•00 元

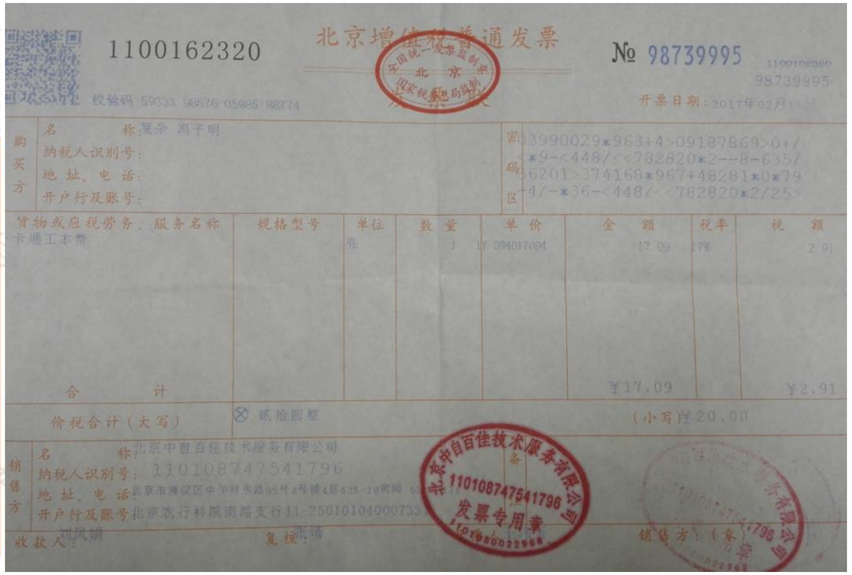
100%


中央非税收入统一票据（电子）
票据代码:00010125
交款人统一社会信用代码:440982********0419
交款人:黄梓纹(黄梓纹)
票据号码:0040976744
校验码:4099a1
开票日期:2025-08-22

项目编码	项目名称	单位	数量	标准	金额（元）	备注
243060	高等学校学费	元	1.00	8,000.00	8,000.00	

金额合计（大写)捌仟元整 (小写)8,000.00
其他信息 黄梓纹
收款单位（章）华南理工大学 复核人: 收款人:zfpt

98.2%


1100162320 北京增值税普通发票 № 98739995

94.3%

➢ Test Case

Document **layout complexity** and **image clarity** directly affect recognition accuracy.
Complex structures (dense tables, irregular formats, stamps/overlaps) increase field misalignment,.
Low clarity (blur, noise, low contrast) causes OCR missing or wrong characters.
As in Fig. 3 (both complex and blurry), these errors can lead to incorrect vouchers and reports.

4 Limitations & Team

Limitation & Team

Challenges

Accounting Correctness

Net / Tax / Total ambiguity

VAT split errors

Balance strictness blocks output

Direction & Templates

Revenue vs Expense misclass

Template coverage incomplete

Payment & Settlement

All paid" assumption

No AP/AR lifecycle

No bank statement matching

Persistence & Traceability

Entries not always persisted

Voucher ↔ invoice linkage weak

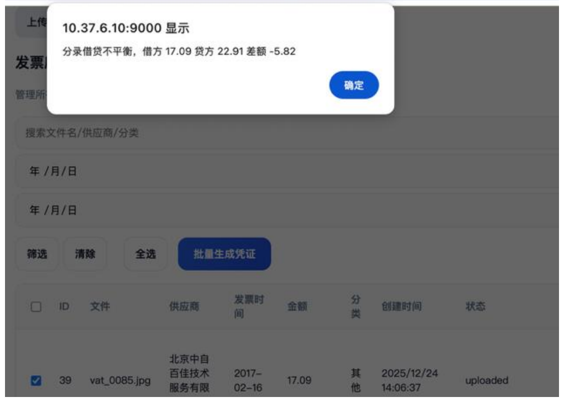
Audit trail not explicit

Future improvement

· **Template & policy engine**: VAT handling, invoice-type detection, deductible rules

· **Robust posting pipeline**: persistence, per-doc error isolation, “needs review” workflow

· **AP/AR + reconciliation**: unpaid tracking, bank statement matching, aging reports

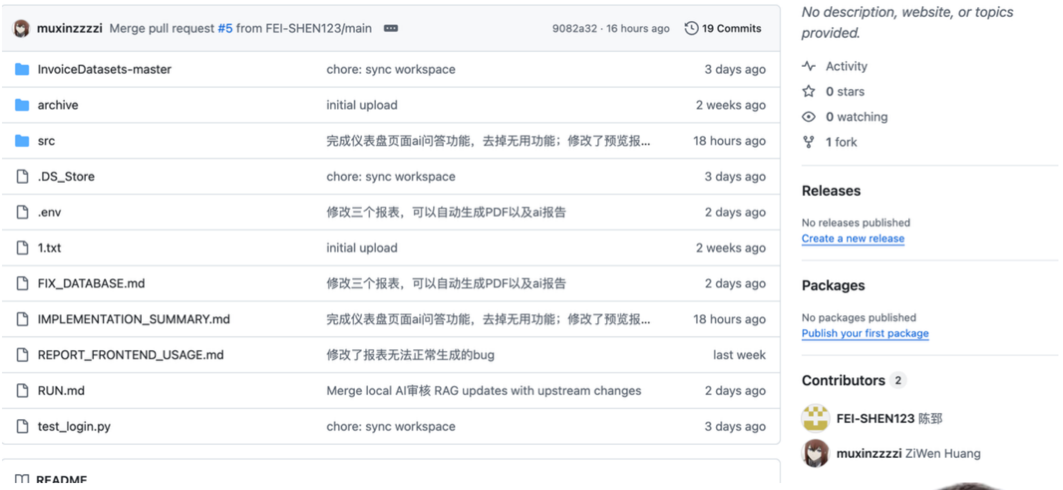


•***Net/Tax/Total mismatch***: *net=17.09 + tax=2.91, but system computed net=14.18 → wrong posting*

Limitation & Team

Team contribution

Huang ziwen	Chen zhi
Core Business Pipeline	Financial Reporting System
Intelligent Validation & Review	AI Assistant
Accounting Module	Users & Access
Intelligent Analytics & Insights	AI Assistant & Q&A System
Data Persistence & Storage	User & Access Management
Frontend Interface	Data Persistence & Storage
API Layer	Voucher Management
Data Processing Capabilities	Security



Thank You!

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